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Higher Secondary Second Year

PTA - Model question paper-I 2020

STD- XII

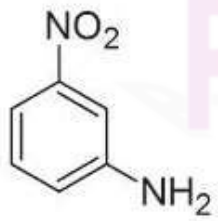
CHEMISTRY

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ANSWER KEY

PART - I

I. Choose the best answer **15 x 1=15**

1. d) $(SiO_4)^{4-}$
2. c) Zone Refining
3. a) HOX > HXO_2 >HXO_3> HXO_4
4. b) 6 Cr-O bonds are equivalent
5. b) CO
6. b) $t = \left(\frac{2}{3}\right)\log 2$
7. a) 74 %
8. a) $PbSO_4$ on cathode is reduced to Pb
9. d) Gel - Butter
10. c) 2 x10^{-3}M
11. b) HCHO > CH_3CHO >C_6H_5CHO >CH_3COCH_3 >$C_6H_5COC_6H_5$
12. b) Both assertion and reason are true but reason is not the correct explanation of assertion
13. c) CH_3-CH_2-CH_2-COCl
14. c) 

15. d) α - helical backbone

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PART

II. Answer any 6 questions (Question no.24 is compulsory) 6 x 2 =12

16. What is the role of depressing agent in froth flotation process ?

When impurities such as ZnS is present in galena (PbS), sodium cyanide (NaCN) is added to depresses the flotation property of ZnS by forming a layer of zinc complex $Na_2[Zn(CN)_4]$ on the surface of zinc sulphide .

17. Why fluorine is more reactive than other halogens?

I) Due to very small size of F there is inter electronic repulsion in F_2

II) This is due to the low value of F -F bond dissociation energy.

iii) Fluorine wants to get to a stable 10 electrons to be like Neon.

iv) This is because the valence/ bonding electrons are closer to the nucleus in Fluorine than they are Chlorine and others and thus more strongly attracted Fluorine is most electronegative, thus it is most reactive.

18. Define solubility product

The solubility product of a compound is defined as the product of the molar concentration of the constituent ions, each raised to the power of its stoichiometric co – efficient in a balanced equilibrium equation.

19. Peptising agent is added to convert precipitate into colloidal solution. Explain with an example.

When peptising agent is added to freshly prepared precipitate, it changes into colloidal sol. It is called peptisation. During peptisation, the precipitate adsorbs one of the ions of the electrolyte on its surface

For example,

when we add a small volume of very dilute hydrochloric acid solution(peptizing agent) to a fresh precipitate of aluminum hydroxide

20. A conductivity cell has two platinum electrodes separated by a distance 1.5 cm and the cross sectional area of each electrode is 4.5 sq cm. Using this cell, the resistance of 0.5 N electrolytic solution was measured as 15Ω . Find the specific conductance of the solution.

Solution

$$l = 1.5\ \text{cm} = 1.5\times10^{-2}\text{m}$$

$$A = 4.5\ \text{cm}^2 = 4.5\times(10^{-4})\text{m}^2$$

$$R = 15\Omega$$

$$\kappa = \frac{1}{R}\left(\frac{l}{A}\right)$$

$$\kappa = \frac{1}{15\Omega}\times\frac{1.5\times10^{-2}\text{m}}{4.5\times10^{-4}\text{m}^2}$$

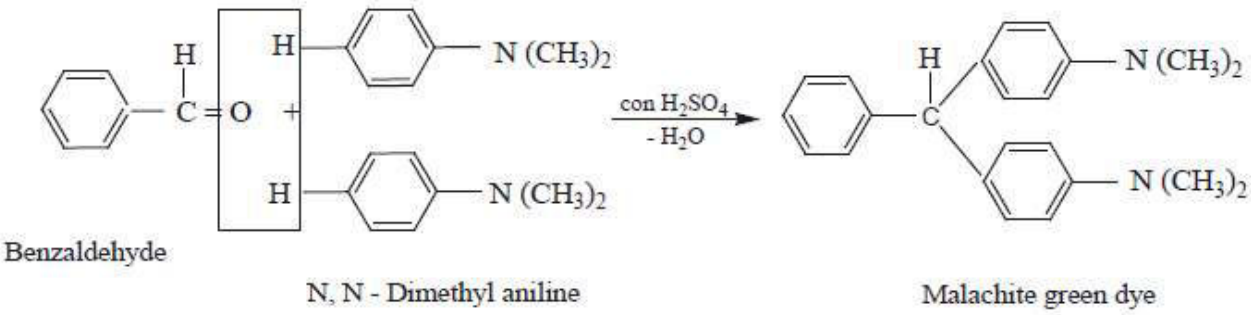
$$= 2.22\ \text{Sm}^{-1}$$

21. Is it possible to oxidise t – butyl alcohol using acidified dichromate to form a carbonyl compound.

i) Tertiary alcohol does not allow for their oxidation because of absence a- hydrogen atom

ii) This is why tertiary alcohols are said to be resistant to oxidation But at elevated temperatures it possible to oxidise, under strong oxidising agent like acidified $K_2Cr_2O_7/\ H^+$ cleavage of C –C bond takes place to give a mixture of carboxylic acid like formic acid and acetic acid.

22. How will you prepare Malachite green dye from benzaldehyde?



Benzaldehyde — N, N - Dimethyl aniline — Malachite green dye

23. i) In decreasing order of the pKb values : $C_2H_5NH_2$, $C_6H_5NHCH_3$, $(C_2H_5)_2NH$ and CH_3NH_2

ii) Increasing order of basic strength : $C_6H_5NH_2$,$C_6H_5N(CH_3)_2$, $C_2H_5)_2NH$ and CH_3NH_2

i) Larger the value of Kb or smaller the value of pKb , stronger is the base.

i) In $C_6H_5NHCH_3$ is less basic than $((C_2H_5)_2NH$ and $C_2H_5NH_2$ due to the delocalization of the lone pair in the former two. Further, among $C_6H_5NHCH_3$ the former will be more basic due to the **+ I** effect of $-CH_3$ group. Hence, the order of increasing basicity of the given compounds is as follows:

$$C_6H_5NHCH_3 < CH_3NH_2 < C_2H_5NH_2 < (C_2H_5)_2NH$$

We know that the higher the basic strength, the lower is the pKb values.

$$C_6H_5NHCH_3 > CH_3NH_2 > C_2H_5NH_2 > (C_2H_5)_2NH$$

ii) Increasing order of basic strength :

ii) $C_6H_5N(CH_3)_2$ is more basic than $C_6H_5NH_2$ due to the presence of the + I effect of two - CH_3 group in $C_6H_5N(CH_3)_2$ Further CH_3NH_2 contains one - CH_3group while $(C_2H_5)_2NH$ contains two - C_2H_5 group. Thus, $(C_2H_5)_2NH$ is more basic than CH_3NH_2 Now $C_6H_5N(CH_3)_2$ is less basic than CH_3NH_2 because of the - R effect of - C_6H_5 group. Hence, the increasing order of the basic strength of the given compounds is as follows

$$C_6H_5NH_2 < C_6H_5N(CH_3)_2 < CH_3NH_2 < (C_2H_5)_2NH$$

24. A hydride of 2nd period alkali metal(X) on reaction with compound of Boron (Y) to give a reducing agent (Z).Identify X, Y and Z.

A hydride of 2nd period alkali metal is - LIH

$$4LiH + BF_3 \rightarrow LiBH_4 + 3LiF$$

X - is lithium hydride (LiH) **Y -** is boron trifluoride (BF_3) **Z -** is lithium borohydride ($LiBH_4$)

Lithium borohydride is used in organic synthesis as a reducing agent for esters.

PART - III

III. Answer any 6 questions(Question no.33 is compulsory) 6 x 3 = 18

25. Show that sulphuric acid is a dehydrating agent, by giving suitable examples.

It is highly soluble in water and has strong affinity towards water and hence it can be used as a dehy drating agent. When dissolved in water, it forms mono ($H_2SO_4.H_2O$) and dihydrates ($H_2SO_4.2H_2O$) and the reaction is exothermic.

$$C_{12}H_{22}O_{11} + H_2SO_4 \rightarrow 12C + H_2SO_4.11H_2O$$

Sulphuric acid reacts with sucrose to produce carbon and hydrated sulphuric acid.

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26. What are interstitial compounds? How they differ from the properties of its pure metals?

An interstitial compound or alloy is a compound that is formed when small atoms like hydrogen, boron, carbon or nitrogen are trapped in the interstitial holes in a metal lattice .

eg) TiC, ZrH , MnN etc

Differ from the properties of its pure metalsis

(i) They are hard and show electrical land thermal conductivity

(ii) They have high melting points higher than those of pure metals

(iii) Transition metal hydrides are used as powerful reducing agents

(iv) Metallic carbides are chemically inert.

27. What is crystal field stabilization energy (CFSE)?

The crystal field stabilisation energy is defined as the energy difference of electronic configurations in theligand filed (E_{LF}) and the isotropic field/ barycentre (E_{iso}).

$$CFSE(\Delta E_o) = \{E_{LF}\} - \{E_{iso}\}$$

$$= \{ [n_{t2g}(-0.4) + n_{eg}(0.6)] \Delta_o + n_pP\} - \{ n'_pP\}$$

nt_{2g} - is the number of electrons in t_{2g} orbitals

n_{eg} - is number of electrons in eg orbitals

n_p is number of electron pairs in the ligand field

n'_p is the number of electron pairs in the isotropic field (barycentre).

28. Differentiate crystalline solids and amorphous solids

S.no	Crystalline solids	Amorphous solids
1	Long range orderly arrangement of constituents.	Short range, random arrangement of constituents.
2	Definite shape	Irregular shape
3	Generally crystalline solids are anisotropic in nature	They are isotropic* like liquids
4	They are true solids	They are considered as pseudo solids (or) super cooled liquids
5	Definite Heat of fusion	Heat of fusion is not definite
6	They have sharp melting points.	Gradually soften over a range of temperature and so can be moulded.
7	Examples: NaCl , diamond etc.,	Examples: Rubber , plastics, glass etc

29. What is the role of adsorption in the heterogeneous catalysis?

Role of adsorption in heterogeneous catalysis-

i) Diffusion of reactants to the surface of the catalyst.

ii) Adsorption of reactant is on the surface of the catalyst.

iii)The chemical reaction can occur at the catalyst surface by intermediate formation between them.

iv) Desorption of the product of the reaction is also at the surface of the catalyst.

30. State and derive Ostwald dilution law.

Ostwald's dilution law relates the dissociation constant of the weak acid (K a) with its degree of dissociation (α) and the concentration (C).

$$\alpha = \frac{\text{Number of moles dissociated}}{\text{total number of moles}}$$

The dissociation of acetic acid can be represented as

$$CH_3COOH \rightleftharpoons H^+ + CH_3COO^-$$

	CH_3COOH	H^+	CH_3COO^-
Initial number of moles	1	-	-
Degree of dissociation	α		
Number of moles at equilibrium	$1-\alpha$	α	α
Equilibrium concentration	$C(1-\alpha)$	$C\alpha$	$C\alpha$

$$Ka = \frac{[H^+][CH_3COO^-]}{[CH_3COOH]}$$

$$K_a = \frac{C\alpha \times C\alpha}{C(1-\alpha)}$$

$$K_a = \frac{C\alpha^2}{1-\alpha}, (1-\alpha \simeq 1)$$

$$K_a = C\alpha^2$$

$$\alpha = \sqrt{\frac{K_a}{C}}$$

$[H+] = C\alpha$ $[H+] = C\sqrt{\frac{K_a}{C}}$ for weak acid is $[H+] = \sqrt{K_a.C}$

31. Name the vitamins whose deficiency cause

i) Pellagra ii) Beri-Beri iii) Night blindness

i) Pellagra (photo sensitive dermatitis)- Vitamin B_3(Niacin)

ii) Beri-Beri - Vitamin B_1(Thiamine)

iii) Night blindness - Vitamin A(Retinol)

32. What are bio degradable polymers? Give two examples.

The materials that are readily decomposed by microorganisms in the environment are called biodegradable

Example i) Polyhydroxy butyrate (PHB)

ii) Polyglycolic acid (PGA),

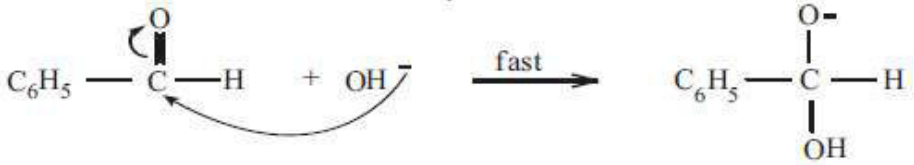
iii) Polylactic acid (PLA)

33. Explain the mechanism of cannizaro reaction

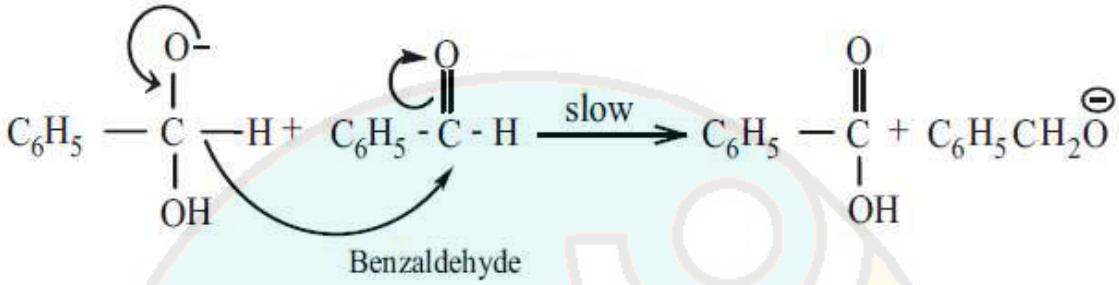
Mechanism of Cannizaro reaction

Cannizaro reaction involves three steps.

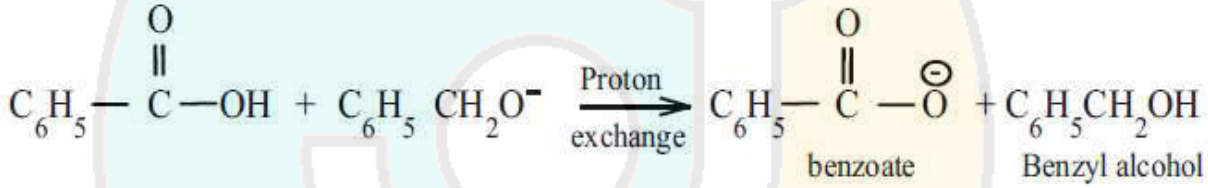
Step 1 : Attack of OH^- on the carbonyl carbon.



Step 2 : Hydride ion transfer



Step 3 : Acid – base reaction.



Cannizaro reaction is a characteristic of aldehyde having no α – hydrogen.

PART - IV

IV . Answer all the questions **5 x 5 = 25**

34. a) Mention the main assumption of valence bond theory of coordination compounds.

1. The ligand . metal bond in a coordination complex is covalent in nature. It is formed by sharing of electrons (provided by the ligands) between the central metal atom and the ligand.
2. Each ligand should have at least one filled orbital containing a lone pair of electrons.
3. In order to accommodate the electron pairs donated by the ligands, the central metal ion present in a complex provides required number (coordination number) of vacant orbitals.
4. These vacant orbitals of central metal atom undergo hybridisation, the process of mixing of atomic orbitals of comparable energy to form equal number of new orbitals called hybridised orbitals with same energy.
5. The vacant hybridised orbitals of the central metal ion, linearly overlap with filled orbitals of the ligands to form coordinate covalent sigma bonds between the metal and the ligand.
6. The hybridised orbitals are directional and their orientation in space gives a definite geometry to the complex ion.

sp - Linear **sp^2 - Trigonal planar**

sp^3 - Tetrahedral **dsp^2 - Square planar**

dsp^3 - Trigonal bipyramidal **sp^3d^2 - Octahedral**

(OR)

b) i) Justify the position of lanthanides and actinoids in the periodic table

The position of Lanthanoids in the periodic table

i) The actual position of Lanthanoids in the periodic table is at group number 3 and period number 6.

However, in the sixth period after lanthanum, the electrons are preferentially filled in inner 4f sub shell and these fourteen elements following lanthanum show similar chemical properties.

ii) Therefore these elements are grouped together and placed at the bottom of the periodic table.

The position of actinoids in the periodic table

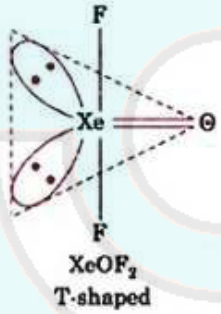
i) Similarly the fourteen elements following actinium resemble in their physical and chemical properties.

ii) If we place these elements after Lanthanum in the periodic table below 4d series, the properties of the elements belongs to a group would be different and it would affect the proper structure of the periodic table.

ii) What is the hybridisation in $XeOF_2$? Give its structure

$XeOF_2$ involves sp^3d hybridization.

The two lone pairs of electrons are at the corners of triangle and the molecule has **T shape**.



35. a) i) Which is known as inorganic benzene? How it is prepared?

When treated with excess ammonia at low temperatures diborane gives diboranediammonate. On heating at higher temperatures it gives borazole

$$3B_2H_6 + 6NH_3 \xrightarrow{-153\,K} 3B_2H_6.2NH_3 \text{ (or) } 3[BH_2(NH_3)_2]^+[BH_4]^-$$

$$3B_2H_6 + 2NH_3 \xrightarrow[\text{Clossed vessel}]{\text{High temp}}$$



$2B_3N_3H_6$ (Borazole or Borazine - Inorganic benzene)

ii) What is the role of graphite rods in the electro metallurgy of Aluminium?

i) In the synthesis of aluminium from electrometallurgically, graphite rod is used as an anode and graphite made line iron is used as cathode.

ii) During the electrolysis process, aluminium is liberated at the cathode and oxygen at the anode.

iii) Oxygen reacts with the graphite anode and forms CO and CO_2.

This prevents the oxygen evolved from oxidizing aluminium formed. Also, for the fact that graphite is the cheapest means of pure carbon, it is advantageous to use it.

(OR)

b) i) Write any three differences between order and molecularity.

	Order of a reaction	Molecularity of a reaction
1.	It is the sum of the powers of concentration species terms involved in the experimentally step.determined rate law.	It is the total number of reactant that are involved in an elementary
2.	It can be zero (or) fractional (or) integer	It is always a whole number, cannot be zero or a fractional number.
3.	It is assigned for a overall reaction.	It is assigned for each elementary step of mechanism.

ii) What are the two types of buffer solution? Give example for each type.

There are two types of buffer solutions.

1. Acidic buffer solution : a solution containing a weak acid and its salt.

Example : solution containing acetic acid and sodium acetate

2. Basic buffer solution : a solution containing a weak base and its salt.

Example : Solution containing NH_4OH and NH_4Cl

36. a) i) What are the conventions used in Galvanic cell notation

The galvanic cell is represented by a cell diagram, for example, Daniel cell is represented as

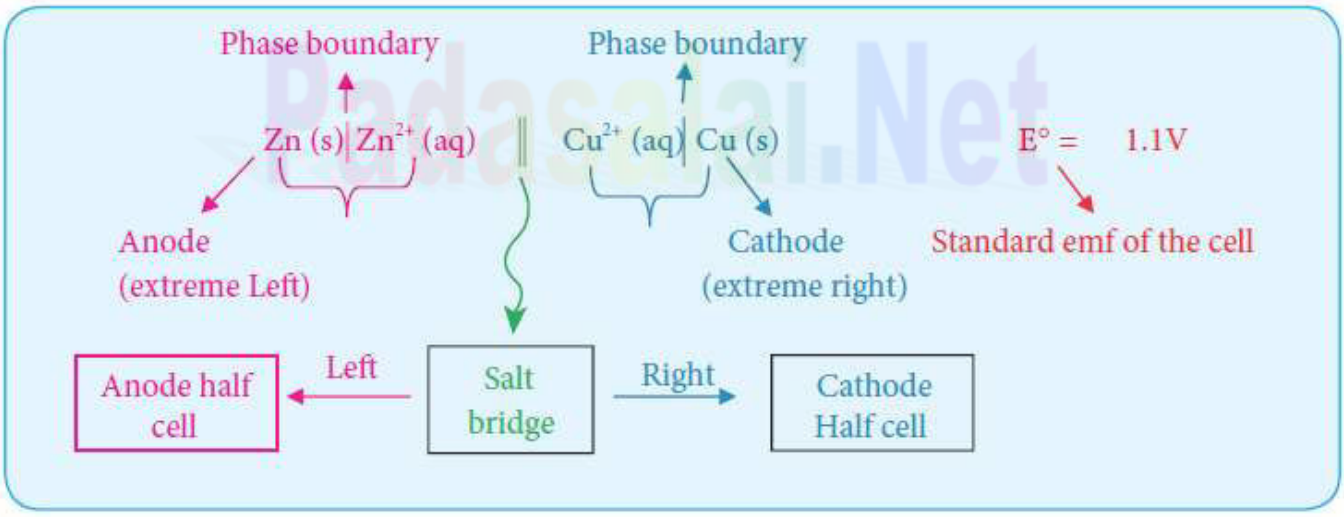
$$Zn_{(s)} \mid Zn^{2+}_{(aq)} \parallel Cu^{2+}_{(aq)} \mid Cu_{(s)}$$

i) In the above notation, a single vertical bar (|) represents a phase boundary and the double vertical bar (||) represents the salt bridge.

ii) The anode half cell is written on the left side of the salt bridge and the cathode half cell on the right side.

iii) The anode and cathode are written on the extreme left and extreme right, respectively.

(OR)



ii) What is intercalation?

In chemistry, intercalation is the reversible inclusion or insertion of a molecule (or ion) into materials with layered structures.

Examples are found in graphite and transition metal dichalcogenides.

(OR)

b) i) Define Emulsification and Deemulsification.

i) The process of preparation of emulsion by the dispersal of one liquid in another liquid is called **Emulsification**

Most of the lyophillic colloids also act as emulsifiers. **Example: glue, gelatine**

ii) Emulsion can be separated into two separate layers. The process is called **Deemulsification**.

Destroying the emulsifier using chemical methods.

ii) Write Debye - Huckel and sagar equation for a uni-univalent electrolyte

For a uni – univalent electrolyte the Debye Huckel and Onsager equation is given below.

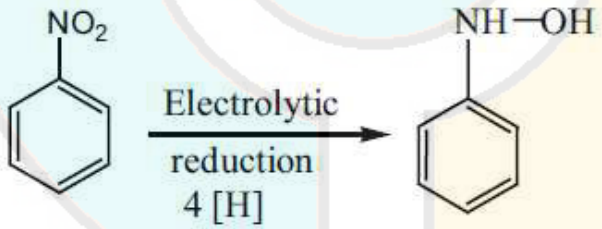
$$\Lambda_m = \Lambda_m^0 - (A + B\Lambda_m^0)\sqrt{C}$$

Where A and B are the constants which depend only on the nature of the solvent and temperature. The expression for A and B are

$$A = \frac{82.4}{\sqrt{DT}\eta} \quad \text{and} \quad B = \frac{8.20 \times 10^5}{\sqrt[3]{DT}}$$

37. a) How the following conversion are effected?

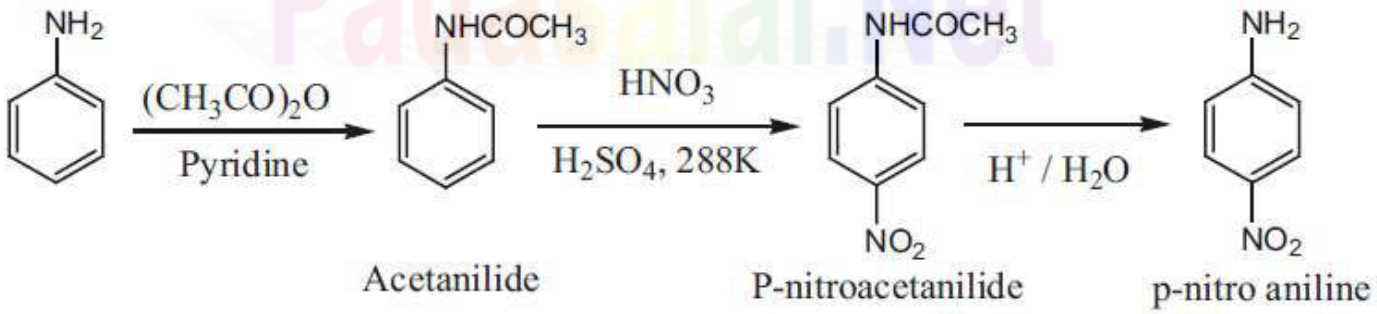
i) Nitro benzene → N- phenyl hydroxyl amine



ii) Propanamide → Propan-1-amine

$$CH_3–CH_2\text{-}CH_2\text{-}CONH_2 + Br_2 + 4\ NaOH \rightarrow CH_3–CH_2\text{-}CH_2\text{-}NH_2 + 2Na_2CO_3 + 2NaBr + 2H_2O.$$

iii) Aniline → p-nitroaniline



(OR)

b) i) What are enzymes?Explain the mechanism of enzyme action.

Th ere are many biochemical reactions that occur in our living cells. Digestion of food and harvesting the energy from them, and synthesis of necessary molecules required for various cellular functions are examples for such reactions. All these reactions are catalysed by special proteins called enzymes.

Mechanism of enzyme action:

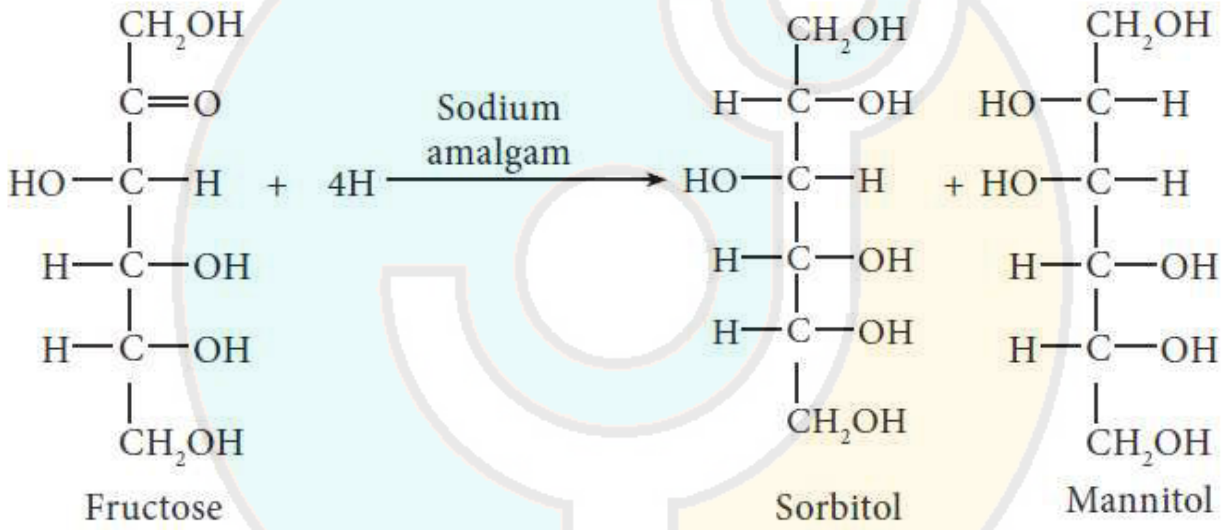
Enzymes are biocatalysts that catalyse a specifi c biochemical reaction. Th ey generally activate the reaction by reducing the activation energy by stabilising the transition state.

$$\underset{\text{Enzyme}}{\mathbf{E}} + \underset{\text{Substate}}{\mathbf{S}} \rightleftharpoons \underset{\text{Enzyme - substrate complex}}{\mathbf{[ES]}}$$

$$\mathbf{[ES]} \rightarrow \mathbf{E + P}$$

ii) What happens when fructose is partially reduced with sodium amalgam and water?

Partial reduction of fructose with sodium amalgam and water produces mixtures of sorbitol and mannitol which are epimers at the second carbon. New asymmetric carbon is formed at C-2. This confims the presence of a keto group.



38. a) i) Atom X and Y from BCC crystalline structure.Atom X is present at the corners of the cube and Y is at the centre of cube. What is the formula of the compound?

$$X = 8\frac{1}{8} = 1(\text{at corner})$$

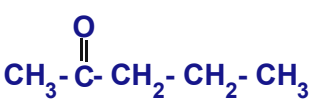
$$Y = 1x1 = 1(\text{at body center})$$

So the formula of compound is =XY.

ii) A carbonyl compound A having molecular formula $C_5H_{10}O$ forms crystalline precipitate with sodium bisulphite and gives positive iodoform test. A does not reduce fehling solution.Identify'A'

Molecular formula of the compound is $C_5H_{10}O$, Since the compound does not reduces Fehling solution Which means it is not an aldehyde, it is a ketone

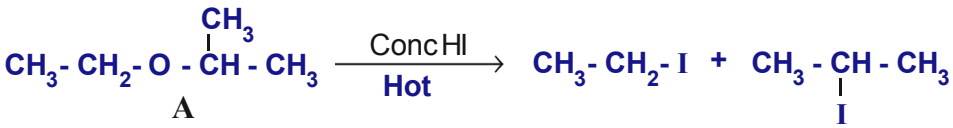
Since the compound gives positive iodoform test it must contain **α- methyl group** . Hence Methyl ketone is **Compound A is 2-pentanone**



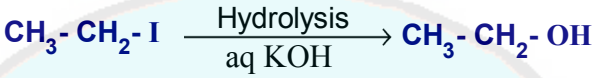
(OR)

b) A ether (A) $C_5H_{12}O$ when heated with excess of hot concentration HI, produced two alkyl halides, which on hydrolysis forms compound (B) and (C).Oxidation of (B) gives an acid (D) where as oxidation of (C) gives ketone (E).Identify A,B,C,D and E and write the chemical equation.

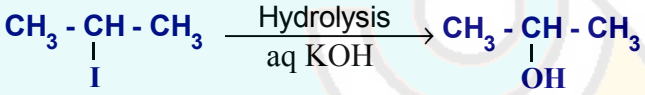
A ether (A) $C_5H_{12}O$ when heated with excess of hot concentration HI, produced two alkyl halides (ethyl iodide and isopropyl iodide)



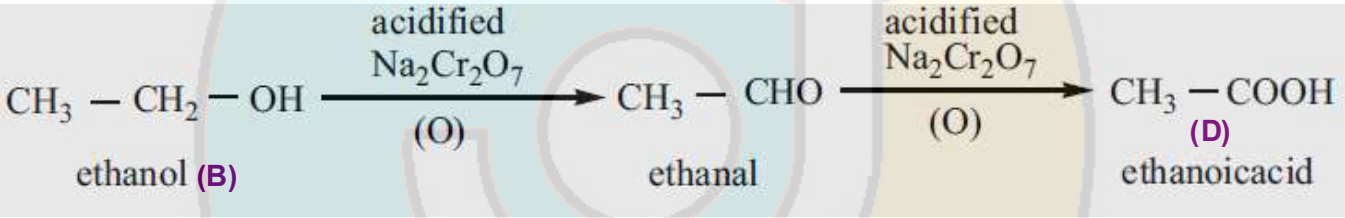
Hydrolysis compound ethyl iodide to give Compound - (B) ethanol



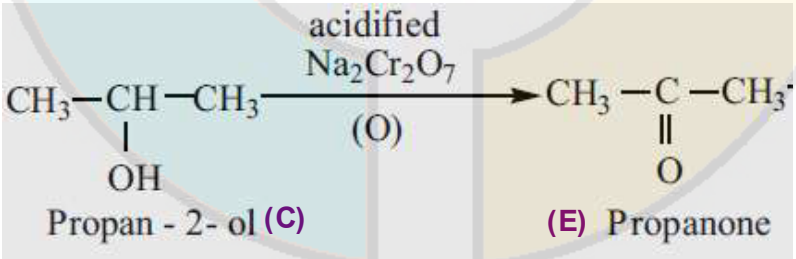
Hydrolysis isopryl iodide to give compound (C) 2-propanol



Compound 'B' on Oxidation to give compound (D) is carboxylic acid



Compound 'C' on Oxidation to give compound (E) is ketone



Compound A - 2-ethoxy propane
Compound B - ethanol
Compound C - 2-propanol (iso propyl alchol)
Compound D - Acetic acid
Compound E - propanone

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